

demo_defireEv2&Ev3 Card

Development manual

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Version	V1.0.0

1. the command return value description

Success: No data: Success (PS: Return value is 0)

There is data: related command requirements data (such as: read data)

Failure: Error: (error code)

2. error code description

Negative error code: (mainly PICC and PCD communication is lost, contact failure. You need to re-check the card)

-2 APK layer interface Introduction parameter error

-10 APK layer transparent error

-1001 parameter error

-1002 IO error

-1003 timeout error

-1004 Check error

-1005 frame error

-1006 interference error

-1007 protocol error

-1008 card conflict error

-1009 send error

16 credit error code: (card instruction operation problem)

The format is 0x13xx, the error code is XX, "DS226031 - Product Data Sheet Mifare Desfire Ev2 Contactless Multi-Application IC (3.1) .pdf" Chapter VII or "Desfire EV3 User MANUAL 3.0.pdf" Chapter 8 finds related commands , Matching this XX information

3. the operating instructions

1. Enter the operating interface and place the card in the identification area.

2. Click "Open NFC", such as opening success, button "Check IC Card" can operate, continue 3.

3. Click "Check IC Card", such as no card, the information bar displays "not retrieved to IC card"; if there is a card, the information bar displays the information, the command button can operate, you can continue to test the relevant command.

4. The first-time EV2 / EV3 card needs to be activated to use, click the button "New Card First USE" to activate.

When the application exits, the NFC is automatically turned off, and can be saheder.

4. Precautions

This Demo is commanded, this parameter is only example, if you need to modify, please refer to Manual "DS226031 - Product Data Sheet Mifare Desfire Ev2 ContactLi-Application IC (3.1) .pdf" or "Desfire EV3 User Manual 3.0.pdf"

5. Operation analysis

Note: Verify an instruction of the EV2 card requires prerequisites, such as AuthenticateEv2First, first need to call SelectApplication, and successfully execute the AuthenticateEv2First to complete the verification. So each command operation in DEMO is integrated, the input parameter is 16, and below is an instructions for each operation in the test interface.

1. AuthenticateEV2First in Demo

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

①

②

① : SelectApplication parameters, select application

② : AuthenticateEv2First parameters, this parameter consists of: Key (16 bytes) + keyno (1 byte) + lencap (1 byte), the default is 0x00

return value:

Success: Success

Failure: Error: (error code)

2. AuthenticateEV2NonFirst in Demo

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

①

②

① : SelectApplication parameters, select application

② : The parameters of Authenticateev2nonfirst, this parameter consists of: key (16 bytes) + Keyno (1 byte), the default is 0x00

return value:

Success: Success

Failure: Error: (error code)

3. FreeMem in Demo

return value:

Success: 3 Bytes data

Failure: Error: (error code)

4. Format in Demo

return value:

Success: Success

Failure: Error: (error code)

00 00 00 00 00

① ②

- ① : SelectApplication parameters, select application
- ② : SetConfiguration parameters, this parameter consists of option (1 byte) + Data (1 byte)

return value:

Success: Success

Failure: Error: (error code)

6. GetVersion in Demo

return value:

Success: 28 Bytes

Failure: Error: (error code)

7. GetCardUID in Demo

return value:

Success: n bytes' data (N is determined by the card, generally 7 bytes)

Failure: Error: (error code)

8. ChangeKey in Demo

[illegible]

①

②

- ① : SelectApplication parameters, select application
- ② : ChangeKey's parameters, composition: ISEqualgetKeyno (1 byte) + KeyNO (1 byte) + NEWKEY (16 Bytes) + Oldkey (16 bytes)

return value:

Success: Success

Failure: Error: (error code)

9. ChangeKeyEV2 in Demo

[illegible]

①②

- ① : SelectApplication parameters, select application

② : ChangeKeyev2 parameters consisting: ISEqualgetKeyno (1 byte) + Keysetno (1 byte) + KeyNO (1 byte) + NEWKEY (16 BYTES) + Oldkey (16 bytes)

return value:

Success: Success

Failure: Error: (error code)

10.InitializeKeySet in Demo

00 00 01 01 02

① ②

①: SelectApplication parameters, select application

②: InitializeKeyset parameters, Composition: KeySetNo (1BYTE) + KeySetType (1byte)

return value:

Success: Success

Failure: Error: (error code)

11.FinalizeKeySet in Demo

00 00 01 01 01

① ②

①: SelectApplication parameters, select application

②: FinalizeKeyset parameters, Composition: KeysetNo (1BYTE) + KeySetVersion (1byte)

return value:

Success: Success

Failure: Error: (error code)

12.RollKeySet in Demo

00 00 01 01

① ②

①: SelectApplication parameters, select application

②: RollKeyset parameters consisting: KeySetNo (1byte)

return value:

Success: Success

Failure: Error: (error code)

13.GetKeySettings in Demo

00 00 00

①

① : SelectApplication parameters, select application

return value:

Success: 2 bytes

Failure: Error: (error code)

14.ChangeKeySettings in Demo

00 00 00 0F

① ②

① : SelectApplication parameters, select application

② : ChangeKeySettings Parameters, KeySettings (1byte)

return value:

Success: Success

Failure: Error: (error code)

15.GetKeyVersion in Demo

00 00 00 00 00 00

① ②

① : SelectApplication parameters, select application

② : GetKeyVersion parameters, Keyno (1 byte) + keysetno (1 byte)

return value:

Success: 1 bytes

Failure: Error: (error code)

16.CreateApplication in Demo

00 00 01 0F 91 01 00 10 10 00

①

①: CreateApplication Parameters, Composition: AID (3 Bytes) + KeySett1 (1 Byte) + KeySett2 (1 Byte) + KeySett3 (1 Byte) + AKSVERSION (1 Byte) + NokeySets (1 Byte) + MaxKeysize (1 Byte) + AppKeySetSett 1 byte)

return value:

Success: Success

Failure: Error: (error code)

17.CreateDelegatedApplication in Demo

00 00 02 00 00 00 10 00 0F 81 00 00 00 00 00

①

①: CreateDelegatedApplication Parameters, Composition: AID(3 bytes) + DAMSlotNo(2 bytes) + DAMSlotVersion(1 byte) + QuotaLimit(2 bytes) + KeySett1(1 byte) + KeySett2(1 byte) +

KeySett3(1 byte) + AKSVersion(1 byte) + NoKeySets(1 byte) + MaxKeySize(1 byte) +
AppKeySetSett(1 byte)

return value:

Success: Success

Failure: Error: (error code)

18.DeleteApplication in Demo

00 00 00 01 00

① ② ③

①: AuthKey certification number, default piccmasterkey 0x00

②: DeleteApplication parameters, Composition: AID (3 bytes)

③: Application certification type, default 0x00, EV3 is used, corresponding to ①

return value:

Success: Success

Failure: Error: (error code)

19.SelectApplication in Demo

00 00 01

①

① : SelectApplication parameters, Composition: AID (3 bytes)

return value:

Success: Success

Failure: Error: (error code)

20.GetApplicationIDs in Demo

return value:

Success: n Bytes (n is determined by the application, one application 3 bytes, M app n = 3 * m bytes)

Failure: Error: (error code)

21.GetDelegatedInfo in Demo

00 00

①

①: GetDelegatedInfo parameters, Composition: DAMSlotNo(2 bytes)

return value:

Success: 8 Bytes (1 Byte) + Quotalimit (2 Bytes) + FreeBlocks (2 Bytes) + AID (3 bytes))

Failure: Error: (error code)

22.CreateStdDataFile in Demo

Ev2 → 00 00 01 01 01 00 00 20 00 00

Ev3 → 00 00 01 01 01 00 00 20 00 00 07 41 00 00 20 00 00

①

②

③

① : SelectApplication parameters, select application

② : CreateStdDataFile parameters, Composition: FileNo(1 byte) + FileOption(1 byte) + AccessRights(2 bytes + FileSize(3 bytes)

③ : CreateStddatafile parameters, Composition: Fileno (1 Byte) + FileOption (1 Byte) + Accessrights (2 Bytes + FileSize (3 bytes), create a file, used for getFileCounters authentication (EV2 does not exist)

return value:

Success: Success

Failure: Error: (error code)

23.CreateBackupDataFile in Demo

00 00 01 02 01 00 00 20 00 00

①

②

①: SelectApplication parameters, select application

②: CreateBackupDataFile parameters, Composition: FileNo(1 byte) + FileOption(1 byte) + AccessRights(2 bytes + FileSize(3 bytes)

return value:

Success: Success

Failure: Error: (error code)

24.CreateValueFile in Demo

00 00 01 03 01 00 00 00 00 00 50 C3 00 00 C8 00 00 00 01

①

②

①: SelectApplication parameters, select application

②: CreateValueFile parameters, Composition: FileNo(1 byte) + FileOption(1 byte) + AccessRights(2 bytes) + LowerLimit(4 bytes) + UpperLimit(4 bytes) + Value(bytes) + LimitedCreditEnabled(1 byte)

return value:

Success: Success

Failure: Error: (error code)

25. CreateLinearRecordFile in Demo

00 00 01 04 01 00 00 20 00 00 10 00 00

①

②

①: SelectApplication parameters, select application

②: CreateLinearRecordFile parameters, Composition: FileNo(1 byte) + FileOption(1 byte) + AccessRights(2 bytes) + RecordSize(3 bytes) + MaxNoOfRecs(3 bytes)

return value:

Success: Success

Failure: Error: (error code)

26. CreateCyclicRecordFile in Demo

00 00 01 05 01 00 00 20 00 00 10 00 00

①

②

①: SelectApplication parameters, select application

②: CreateCyclicRecordFile parameters, Composition: FileNo(1 byte) + FileOption(1 byte) + AccessRights(2 bytes) + RecordSize(3 bytes) + MaxNoOfRecs(3 bytes)

return value:

Success: Success

Failure: Error: (error code)

27. CreateTransactionMACFile in Demo

00 00 01 06 03 0F FF 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

①

②

①: SelectApplication parameters, select application

②: CreateTransactionMACFile parameters, Composition: FileNo(1 byte) + FileOption(1 byte) + AccessRights(2 bytes) + TMKeyOption(1 byte) + TMKey(16 bytes) + TMKeyVer(1 byte)

return value:

Success: Success

Failure: Error: (error code)

28. DeleteFile in Demo

00 00 01 05

①

②

①: SelectApplication parameters, select application

②: DeleteFile parameters, Composition: FileNo(1 byte)

return value:

Success: Success

Failure: Error: (error code)

29. GetFileIDs in Demo

00 00 01

①

①: SelectApplication parameters, select application

return value:

Success: n bytes (n is determined by the number of files existing in this app, one file 1 bytes, n file n bytes)

Failure: Error: (error code)

30. GetFileSettings in Demo

00 00 01 01

①

②

①: SelectApplication parameters, select application

②: GetFileSettings parameters, Composition:FileNo(1 byte)

return value:

Success: n bytes (n is determined by the file type)

Failure: Error: (error code)

31. ChangeFileSettings in Demo

Ev2→ 00 00 01 01 01 00 00

Ev3→ 00 00 01 07 43 00 00 41 FE FF

①

②

①: SelectApplication parameters, select application

②: ChangeFileSettings Parameters, EV2 Composition: Fileno (1 Byte) + FileOption (1 Byte) + Accessrights (2 Bytes), EV3 Composition: Fileno (1 Byte) + FileOption (1 Byte) + Accessrights (2 Bytes) + SDMOption (1 Byte + SDMACCESSRIGHTS (2 Bytes)

return value:

Success: Success

Failure: Error: (error code)

32. GetFileCounters(only on Ev3) in Demo

00 00 01 07

①

②

①: SelectApplication parameters, select application
②: GetFileCounters parameters, Composition:FileNo(1 byte)
return value:
Success: 5 Bytes (SDMReadctr (3 bytes) + reserved (2 bytes))
Failure: Error: (error code)

33. ReadData in Demo

00 00 01 01 01 00 00 00 15 00 00

① ②

①: SelectApplication parameters, select application
②: ReadData parameters, Composition:CommMode(1 byte) + FileNo(1 byte) + Offset(3 bytes) + Length(3 bytes)
return value:
Success: n bytes (N is determined by the input parameter length, the value of Length is equal to N)
Failure: Error: (error code)

34. WriteData in Demo

00 00 01 01 01 00 00 00 09 00 00 01 02 03 04 05 06 07 08 09

① ②

①: SelectApplication parameters, select application
②: WriteData parameters, Composition: CommMode (1 Byte) + Fileno (1 Byte) + LENGTH (3 Bytes) + Data (consistent with Length)
return value:
Success: Success
Failure: Error: (error code)

35. GetValue in Demo

00 00 01 01 03

① ②

①: SelectApplication parameters, select application
②: GetValue parameters, Composition:CommMode(1 byte) + FileNo(1 byte)
return value:
Success: 4 bytes
Failure: Error: (error code)

36. Credit in Demo

00 00 01 01 03 02 00 00 00

①

②

①: SelectApplication parameters, select application

②: Credit parameters, Composition:CommMode(1 byte) + FileNo(1 byte) + Data(4 bytes)

return value:

Success: Success

Failure: Error: (error code)

37. Debit in Demo

00 00 01 01 03 02 00 00 00

①

②

①: SelectApplication parameters, select application

②: Debit parameters, Composition:CommMode(1 byte) + FileNo(1 byte) + Data(4 bytes)

return value:

Success: Success

Failure: Error: (error code)

38. LimitedCredit in Demo

00 00 01 01 03 02 00 00 00

①

②

①: SelectApplication parameters, select application

②: LimitedCredit parameters, Composition:CommMode(1 byte) + FileNo(1 byte) + Data(4 bytes)

return value:

Success: Success

Failure: Error: (error code)

39. ReadRecords in Demo

00 00 01 01 04 00 00 00 01 00 00

①

②

①: SelectApplication parameters, select application

②: ReadRecords parameters, Composition:CommMode(1 byte) + FileNo(1 byte) + RecNo(3 bytes) + RecCount(3 bytes)

return value:

Success: n bytes (n is determined by the record size set when the loop file is created, namely n = recordsize)

Failure: Error: (error code)

40. WriteRecord in Demo

00 00 01 01 04 00 00 00 01 00 00 01

①

②

①: SelectApplication parameters, select application

②: WriteRecord's parameters, Composition: CommMode (1 Byte) + Fileno (1 Byte) + LENGTH (3 bytes) + Data (consistent with Length)

return value:

Success: Success

Failure: Error: (error code)

41. UpdateRecord in Demo

00 00 01 01 04 00 00 00 00 00 00 01 00 00 01

①

②

①: SelectApplication parameters, select application

②: UpdateRecord Parameters, Composition: CommMode (1 Byte) + Fileno (1 Byte) + Recno (3 Bytes) + LENGTH (3 Bytes) + Data (consistent with Length)

return value:

Success: Success

Failure: Error: (error code)

42. ClearRecordFile in Demo

00 00 01 04

①

②

①: SelectApplication parameters, select application

②: ClearRecordfile parameters, Fileno (1 byte)

return value:

Success: Success

Failure: Error: (error code)

43. CommitTransaction in Demo

01

①

①: CommitTransaction Parameters, Composition: Option(1 byte)

return value:

Success: 12 Bytes (TMC (4 Bytes) + TMV (8 bytes))

Failure: Error: (error code)

44.AbortTransaction in Demo

return value:

Success: Success

Failure: Error: (error code)

45.CommitReaderID in Demo

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

①

①: CommitReaderID Parameters, Composition:TMRI (16 bytes)

return value:

Success: 16 Bytes (EnctMri)

Failure: Error: (error code)

46.Read_Sig in Demo

return value:

Success: 56 Bytes

Failure: Error: (error code)

6. Android API interface

Note: Byte sorting default small-end mode

1. int NFC_Device_Open();

Function: Open NFC

Return value: 0 →success

Non-zero→ failure

2. int NFC_Device_Close();

Function: Turn off NFC

Return value: 0 →success

Non-zero → failure

3. int NFC_Device_Activate(byte[] timeoutMs, byte[] outData, int[] outLen);

Function: Search and activate cards

Parameter: timeoutMs → Timeout 4 byte small end mode

EXAMPLE: 0x10 0x00 0x00 0x00

Established parameters: outData → receives data

outLen → Length of outData

Return value: 0 → success

Non-zero → failure

4. int NFC_Device_Remove();

Function: Remove the card

Return value: 0 → success

Non-zero → failure

5. int CPU_GET_ATS(byte[] outData, int[] outLen);

Function: Get Card ATS

Parameter: no

Established parameters: outData → receives data

outlen → Length of outData

Return: value: 0 → success

Non-zero → failure

6. int AuthenticateISO(byte KeyNo, byte KeyType, byte[] Key);

Function: EV1 card certification, used to activate EV2 / EV3 card

Parameter: KeyNo → 1 byte default 0x00

KeyType → 1 byte default 0x00

Key → 16 byte default is 0x00

Outlet parameters: no

Return value: 0 → success

Non-zero → failure

7. int AuthenticateEV2First(byte[] Key, byte KeyNo, byte LenCap);

Function: EV2 / EV3 card certification

Parameter: KEY → 16 byte authentication, white card default is 0x00

KeyNo → 1 byte

KeyNo	1	-	Targeted authentication key
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6		RFU
	Bit 5-0		Key Number
		Limited range	[if AID 0x000000 selected] At PICC level, key numbers are specified in PICCKeys Table. This table also indicates which keys are available for authentication.
		0x0..0xD	[else] At application level, maximal key number depends on the number of application keys (max 14)

LenCap → 1 byte default is 0x00

Outlet parameters: no

Return value: 0 → success

Non-zero → failure

8. int AuthenticateEV2First(byte[] Key,byte KeyNo,byte

LenCap,byte[]PcdCap2);

Function: EV2 / EV3 card certification

Parameter: KEY → 1616 byte authentication, white card default is 0x00

KeyNo → 16 1 byte

KeyNo	1	-	Targeted authentication key
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6		RFU
	Bit 5-0		Key Number
		Limited range	[if AID 0x000000 selected] At PICC level, key numbers are specified in PICCKeys Table. This table also indicates which keys are available for authentication.
		0x0..0xD	[else] At application level, maximal key number depends on the number of application keys (max 14)

LenCap → 1 byte is not 0x00 value

LenCap	1	0x0..0x6	Length of the PCD Capabilities.
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PcdCap2 → 1 to 6 bytes

PcdCap2	[1..6]	Full range	Capability vector of the PCD.
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Outlet parameters: no

Return value: 0 → success

Non-zero → failure

9. int AuthenticateEV2NonFirst(byte[] Key,byte KeyNo);

Function: EV2 / EV3 card certification

Parameter: KEY → 16 byte authentication, white card default is 0x00

KeyNo → 1 byte

KeyNo	1	-	Targeted authentication key
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6		RFU
	Bit 5-0		Key Number
		Limited range	[if AID 0x000000 selected] At PICC level, key numbers are specified in PICCKeys Table. This table also indicates which keys are available for authentication.
		0x0..0xD	[else] At application level, maximal key number depends on the number of application keys (max 14)

Outlet parameters: no

Return value: 0 → success

Non-zero → failure

10.int FreeMem(byte[] outData,int[] outLen) ;

Function: Get free memory

Parameter: no

Established parameters: outData → receives data

outLen → Length of outData

Return value: 0 → success

Non-zero → failure

11.int Format();

Function: format

Return value: 0 → success

Non-zero → failure

12.int SetConfiguration(byte Option,byte[] Data);

Function: Configuring card properties

Parameter: Option1 byte

Option	1	-	Configuration Option. Define length and content of the Data parameter.
		0x00	PICC Configuration.
		0x01	Default Keys Update.
		0x02	ATS Update.
		0x03	SAK Update
		0x04	Secure Messaging Configuration.
		0x05	Capability Data.
		0x06	VC Installation Identifier
		0x07..0xFF	RFU

Data→1-25bytes

Data	1..26	-	Data content depends on option values.
		Full range	Data content depends on option value as defined in setConfigOptionsList Table.

Outlet parameters: no

Return value: 0→ success

Non-zero→ failure

13.int GetVersion(byte[] outData,int[] outLen);

Function: Get the card version information

Parameter: no

Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

14.int GetCardUID(byte[] outData,int[] outLen);

Function: Get the card UID

Parameter: no

Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→success

Non-zero→ failure

15.int ChangeKeyEv1(byte KeyNo, byte[] Key);

Function: EV1 card modified the secret key for activating EV2 / EV3

Parameter: Keyno1 Byte

KeyNo	1	-	Key number of the key to be changed. If current AID is 0x000000, also the new key type is given by this parameter.
	Bit 7-6		[if AID 0x000000 is selected and targeting KeyID.PICCMasterKey] Key type
		00b	KeyType.2TDEA
		01b	KeyType.3TDEA
		10b	KeyType.AES
		11b	RFU
	Bit 7		[else] Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6		[else] RFU
	Bit 5-0		Key Number
		Limited range	[if AID 0x000000 selected] At PICC level, key numbers are specified in PICCKeys Table.
		0x0..0xD	[if AID 0x000000 is not selected] At application level, maximal key number depends on the number of application keys (max 14)
		0x21..0x23	[if AID 0x000000 is not selected] At application level, these VC keys can be enabled at application creation

Key→116 byte The key after modification, it is set, do not understand, set all 0x00

Outlet parameters: no

Return value: 0→1 success

Non-zero→1 failure

16.int ChangeKey(byte isEqualGetKeyNo,byte KeyNo,byte KeyVer,byte[]

NewKey,byte[] OldKey);

Function: Modify the secret key, or enable other secret keys

Parameter: isEqualGetKeyNo→1 Byte 0x01 or 0x00

Set 0x01 if the certified secret is the same as the target secret key

Set 0x00 if the authentication secret is different from the key number of the target

For example, AuthenticateEV2First or AuthenticateEV2NonFirst's Key number is the same as Key number used by ChangeKey, isEqualGetKeyNo is 0x01, the opposite 0x00

KeyNo→1 byte

KeyNo	1	-	Key number of the key to be changed. If current AID is 0x000000, also the new key type is given by this parameter.
	Bit 7-6		[if AID 0x000000 is selected and targeting KeyID.PICCMasterKey] Key type
		00b	KeyType.2TDEA
		01b	KeyType.3TDEA
		10b	KeyType.AES
		11b	RFU
	Bit 7		[else] Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6		[else] RFU
	Bit 5-0		Key Number
		Limited range	[if AID 0x000000 selected] At PICC level, key numbers are specified in PICCKeys Table.
		0x0..0xD	[if AID 0x000000 is not selected] At application level, maximal key number depends on the number of application keys (max 14)
		0x21..0x23	[if AID 0x000000 is not selected] At application level, these VC keys can be enabled at application creation

KeyVer→ byte is used for new secrets of new secrets, set it it yourself, don't understand 0x00

NewKey→16 bytes, new secret keys, self-setting, do not understand 0x00

OldKey→16 bytes, old secret keys, default all 0x00

Outlet parameters: no

Return value: 0 →success

Non-zero→ failure

17.int ChangeKeyEV2(byte isEqualGetKeyNo,byte KeySetNo,byte

KeyNo,byte KeyVer,byte[] NewKey,byte[] OldKey);

Function: Modify the secret key, or enable other secret keys

Parameter: isEqualGetKeyNo→1 Byte 0x01 or 0x00

Set 0x01 if the certified secret is the same as the target secret key

Set 0x00 if the authentication secret is different from the key number of the target

For example, AuthenticateEV2First or AuthenticateEV2NonFirst's Key numberis the same as Key number used by ChangeKey, isEqualGetKeyNo is 0x01, the opposite 0x00

KeySetNo→1 bytes

KeySetNo	1	-	Key set number
	Bit 7-4		RFU
	Bit 3-0		Key set number within targeted application
		0x0	[if AID 0x000000 is selected] No key sets at PICC level.
		0x0..0xF	[else] Specified key set containing the targeted key

KeyNo→1 bytes

KeyNo	1	-	Key number of the key to be changed. If current AID is 0x000000, also the new key type is given by this parameter.
	Bit 7-6		[if AID 0x000000 is selected and targeting KeyID.PICCMasterKey] Key type
		00b	KeyType.2TDEA
		01b	KeyType.3TDEA
		10b	KeyType.AES
		11b	RFU
	Bit 7		[else] Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6		[else] RFU
	Bit 5-0		Key Number
		Limited range	[if AID 0x000000 selected] At PICC level, key numbers are specified in PICCKeys Table.
		0x0..0xD	[if AID 0x000000 is not selected] At application level, maximal key number depends on the number of application keys (max 14)
		0x21..0x23	[if AID 0x000000 is not selected] At application level, these VC keys can be enabled at application creation

KeyVer→1 byte is used for new secret keys, set up, set 0x00 if you don't understand

NewKey→16bytes, new secret keys, settled, set 0x00 if you don't understand

OldKey→16 bytes, old secret keys, default all 0x00

Outlet parameters: no

Return value: 0 →success

Non-zero→ failure

18.int InitializeKeySet(byte KeySetNo,byte KeySetType);

Function: Initialization Secret Key Set

Parameter: KeySetNo→1 byte

KeySetType→1 byte

KeySetNo	1	-	Key set number
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-4		RFU
	Bit 3-0		Key set number within targeted application
		0x1..0xF	
KeySetType	1	-	Key type of the key set to be initialized.
	Bit 7-2		RFU
	Bit 1-0		Key type of the application keys
		00b	KeyType.2TDEA
		01b	KeyType.3TDEA
		10b	KeyType.AES
		11b	RFU

Outlet parameters: no

Return value: 0 → success

Non-zero → failure

19.int FinalizeKeySet(byte KeySetNo,byte KeySetVersion);

Function: Complete the secret key set

Parameter: KeySetNo → 1 byte

KeySetVersion → 1 bytes

KeySetNo	1	-	Key set number
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-4		RFU
	Bit 3-0		Key set number within targeted application
		0x1..0xF	
KeySetVersion	1	0x01..0xFF	Key set version of the key set to be finalized.

Outlet parameters: no

Return value: 0 → success

Non-zero → failure

20.int RollKeySet(byte KeySetNo);

Function: Rolling Secret Key Set

Parameter: KeySetNo → 1 byte

Function: Rolling Secret Key Set

Parameter: KeySetNo → 1 byte

KeySetNo	1	-	Key set number
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-4		RFU
	Bit 3-0		Key set number within targeted application
		0x1..0xF	

Outlet parameters: no

Return value: 0→ success

Non-zero→failure

21.int GetKeySettings(byte[] outData,int[] outLen);

Function: Get the secret key settings

Parameter: no

Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero →failure

22.int ChangeKeySettings(byte KeySetting);

Function: Change the key setting

Parameter:KeySetting→1byte

KeySetting	1	-	
		Full range	[if AID 0x000000 selected] PICCKeySettings, see PICCKeySettings Table
		Full range	[else] AppKeySettings, see AppKeySetting Table

Outlet parameters: no

Return value: 0→ success

Non-zero →failure

23.int GetKeyVersion(byte KeyNo, byte KeySetNo, byte[] outData, int[] outLen);

Function: Get the secret key version

Parameter: KeyNo→1 bytes

KeyNo	1	-	Key number of the targeted key
	Bit 7-6		[if AID 0x000000 is selected] RFU
	Bit 7		[else] Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6		[else] KeySetNo presence
		0b	Not present
		1b	Present
	Bit 5-0		Key number of the targeted key
		Restricted Range	[if AID 0x000000 is selected] At PICC level, key numbers are specified in PICCKeys Table
		0x0	[if KeyNo[b6] and KeySetNo[b7] is set] RFU (Key set versions retrieval)
		0x0..0xD	[else] At application level, maximal key number depends on the number of application keys (max 14)
		0x21..23	[else] KeyID.VCKeys (if available at application level)

KeySetNo→1 byte

KeySetNo	[1]	-	[Optional, present if KeyNo[b6] is set] Key set number
	Bit 7		Retrieval Target
		0b	Key version retrieval from specific key set
		1b	Key set versions retrieval
	Bit 6-4		RFU
	Bit 3-0		Key set number within targeted application
		0x0	[if KeySetNo[b7] is set] RFU
		0x0..0xF	[if KeySetNo[b7] is not set] Specified key set containing the targeted key

Outlet parameters: outData→Receive data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

24.int CreateApplication(byte[] AID, byte KeySet1, byte KeySet2, byte KeySet3, byte AKSVersion, byte NoKeySets, byte MaxKeySize, byte AppKeySetSett);

Function: Create an app

Parameter: AID →3 byte

KeySet1→1 bytes

KeySet2→1 bytes

AID	3	-	Application Identifier of the application to be created
		0x000001 .. 0xFFFFF	AID 0x000000 is reserved for the PICC level
KeySet1	1	-	Application Key settings, see Table AppKeySetting.
KeySet2	1	-	Several other key settings
	Bit 7-6		KeyType of the application keys of the initial AKS
		00b	KeyType.2TDEA
		01b	KeyType.3TDEA
		10b	KeyType.AES
		11b	RFU
	Bit 5		Use of 2 byte ISO/IEC 7816-4 File Identifiers
		0b	No 2 byte File Identifiers for files within the application
		1b	2 byte File Identifiers for files within the application required
	Bit 4		KeySet3 presence
		0b	disabled
		1b	enabled
	Bit 3-0		Number of application keys (n)
		0x0..0xE	Maximum 14 keys

KeySet3→1 byte

AKSVersion →1 byte

NoKeySets →1 byte

MaxKeySize →1 byte

AppKeySetSett →1 byte

KeySet3	[1]	-	[Optional, present if KeySet2[b4] is set] Additional optional key settings
	Bit 7-3		RFU
	Bit 2		Application specific Capability data
		0b	disabled
		1b	enabled
	Bit 1		Application specific VC keys
		0b	disabled
		1b	enabled
	Bit 0		Application key sets
		0b	disabled
		1b	enabled
AKSVersion	[1]	Full range	[Optional, present if KeySet3[b0] is set] Key Set Version of the Active Key Set
NoKeySets	[1]	-	[Optional, present if KeySet3[b0] is set] Number of Key Sets
		0x02..0x10	Minimum 2 and maximum 16 key sets
MaxKeySize	[1]	-	[Optional, present if KeySet3[b0] is set] Max. Key Size Has to allow for the specified crypto method in KeySet2 Bit7-6
		0x10	KeyType.2TDEA and KeyType.AES allowed (16 byte keys)
		0x18	All key types allowed (up to 24 byte keys)
AppKeySetSett	[1]	-	[Optional, present if KeySet3[b0] is set] Application Key Set Setting, see AppKeySetSetting Table

Outlet parameters: no

Return value: 0 →success

Non-zero→ failure

25.int CreateDelegatedApplication(byte[] AID, byte[] DAMSlotNo, byte DAMSlotVersion, byte[] QuotaLimit, byte KeySett1, byte KeySett2, byte KeySett3, byte AKSVersion, byte NoKeySets, byte MaxKeySize, byte AppKeySetSett);

Function: Create a delegated application

Introducing parameters:AID →3bytes

DAMSlotNo →2bytes

DAMSlotVersion →1byte

QuotaLimit →2bytes

KeySett1→1byte

KeySett2→1byte

KeySett3→1byte

AKSVersion →1byte

AID	3	-	Application Identifier of the application to be created
		0x000001 .. 0xFFFFFFFF	AID 0x000000 is reserved for the PICC level
DAMSlotNo	2	Full range	Slot number associated with the memory space of the application to be created.
DAMSlotVersion	1	Full range	Slot version associated with the memory space of the application to be created.
QuotaLimit	2	Full range	Maximal memory consumption of the application in 32 byte blocks.
KeySett1	1	-	Application Key settings, see AppKeySetting Table.
KeySett2	1	-	Several other key settings. For detailed definition, see Cmd.CreateApplication.
KeySett3	[1]	-	[Optional, present if KeySett2[b4] is set] Additional optional key settings
AKSVersion	[1]	Full range	[Optional, present if KeySett3[b0] is set] Key Set Version of the Active Key Set

NoKeySets →1byte

MaxKeySize →1byte

AppKeySetSett →1byte

NoKeySets	[1]	-	[Optional, present if KeySett3[b0] is set] Number of Key Sets
		0x02..0x10	Minimum 2 and maximum 16 key sets
MaxKeySize	[1]	-	[Optional, present if KeySett3[b0] is set] Max. Key Size Has to allow for the specified crypto method in KeySett2 Bit7-6
		0x10	KeyType.2TDEA and KeyType.AES allowed (16 byte keys)
		0x18	All key types allowed (up to 24 byte keys)
AppKeySetSett	[1]	-	[Optional, present if KeySett3[b0] is set] Application Key Set Setting, see AppKeySetSetting Table

Outlet parameters: no

Return value: 0 →success

Non-zero→ failure

26.int DeleteApplication(byte[] AID, byte AppType);

Function: Delete an application

Introducing parameters:AID →3byte

AID	3	-	Application Identifier of the application to be deleted
		0x000001 .. 0xFFFFFFFF	AID 0x000000 is reserved for the PICC level and cannot be deleted

AppType→1byte

If the authentication key is PICCMasterKey, AppType is set to 0x00 (applicable EV2 / EV3)

If the authentication key is PICCDAMAuthKey, AppType is set to 0x10 (Applicable EV3)

Outlet parameters: no

Return value: 0→ success

Non-zero→ failure

27.int SelectApplication(byte[] AID);

Function: Select an application

Introducing parameters:AID→3bytes

AID	3	-	Application Identifier of the application to be deleted
		0x000001 .. 0xFFFFFFFF	AID 0x000000 is reserved for the PICC level and cannot be deleted

Outlet parameters: no

Return value: 0 →success

Non-zero→ failure

28.int SelectApplication(byte[] AID,byte[] AID2);

Function: Select 2 applications

Parameter: AID →3 byte

AID2 →3 bytes

AID1	3	Full range	Application Identifier of the primary application to be selected.
AID2	[3]	Full range	[Optional] Application ID of the secondary application to be selected.

Outlet parameters: no

Return value: 0→ success

Non-zero→ failure

29.int GetApplicationIDs(byte[] OutData, int[] OutLen);

Function: Select Apply ID

Parameter: no

Established parameters:outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

30.int GetDelegatedInfo(byte[] DAMSlotNo, byte[] OutData, int[]

OutLen);

Function: Get Delegated application information

Parameter: DAMSlotNo→2 byte

DAMSlotNo	2	Full range	Slot number associated with the memory space of the targeted delegated application
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Established parameters:outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

31.int CreateStdDataFile(byte FileNo,byte FileOption,byte[]

AccessRights,byte[] FileSize);

Function: Create a standard file

Parameter: FileNo → 1byte

FileNo	1	-	File number of the file to be created.
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5		RFU
	Bit 4-0		File number

FileOption →1byte

Ev2:

FileOption	1	-	Options for the targeted file.
	Bit 7		Additional Access Rights
		0b	disabled
		1b	enabled
	Bit 6-2		RFU
	Bit 1-0		CommMode (see Table CommunicationModes)
		x0"	CommMode.Plain
		01b	CommMode.MAC
		11b	CommMode.Full

Ev3:

FileOption	1	-	Options for the targeted file.
	Bit 7		Additional Access Rights
		0b	disabled
		1b	enabled
	Bit 6	-	Secure Dynamic Messaging and Mirroring
		0b	disabled
		1b	enabled
	Bit 5-2	0000b	RFU
	Bit 1-0	-	CommMode (see Table CommunicationModes)
		x0b	CommMode.Plain
		01b	CommMode.MAC
		11b	CommMode.Full

AccessRights→2bytes

AccessRights	2	-	Set of access conditions for the 1st set in the file (see Setaccessconditions Table).
--------------	---	---	---

FileSize →3bytes

FileSize	3	-	File size in bytes for the file to be created.
		0x00001 .. 0xFFFFF	Empty file not allowed.

Outlet parameters: no

Return value: 0 →success

Non-zero→ failure

32.int CreateBackupDataFile(byte FileNo, byte FileOption, byte[]

AccessRights, byte[] FileSize);

Function: Create a backup file

Parameter: FileNo →1byte

FileNo	1	-	File number of the file to be created.
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5		RFU
	Bit 4-0		File number

FileOption →1byte

FileOption	1	-	Options for the targeted file.
	Bit 7		Additional Access Rights
		0b	disabled
		1b	enabled
	Bit 6-2		RFU
	Bit 1-0		CommMode (see Table CommunicationModes)
		x0"	CommMode.Plain
		01b	CommMode.MAC
		11b	CommMode.Full

AccessRights →2bytes

AccessRights	2	-	Set of access conditions for the 1st set in the file (see Setaccessconditions Table).
--------------	---	---	---

FileSize → 3bytes

FileSize	3	-	File size in bytes for the file to be created.
		0x00001 .. 0xFFFFF	Empty file not allowed.

Outlet parameters: no

Return value: 0 → success

Non-zero → failure

33.int CreateValueFile(byte FileNo,byte FileOption,byte[]

AccessRights,byte[] LowerLimit,byte[] UpperLimit,byte[] Value,byte LimitedCreditEnabled);

Function: Create a value file

Parameter: FileNo → 1 byte

FileNo	1	-	File number of the file to be created.
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5		RFU
	Bit 4-0		File number

FileOption → 1byte

FileOption	1	-	Options for the targeted file.
	Bit 7		Additional Access Rights
		0b	disabled
		1b	enabled
	Bit 6-2		RFU
	Bit 1-0		CommMode (see Table CommunicationModes)
		x0"	CommMode.Plain
		01b	CommMode.MAC
		11b	CommMode.Full

AccessRights → 2bytes

AccessRights	2	-	Set of access conditions for the 1st set in the file (see Setaccessconditions Table).
--------------	---	---	---

LowerLimit → 4bytes

LowerLimit	4	-	Lower Limit of the file encoded as a 4 byte signed integer.
		Full Range	with LowerLimit not strictly bigger than UpperLimit

UpperLimit → 4bytes

UpperLimit	4	-	Upper Limit of the file encoded as a 4 byte signed integer.
		Full Range	with LowerLimit not strictly bigger than UpperLimit

Value → 4bytes

Value	4	-	Current Value.
		Full Range	with Value not strictly smaller than LowerLimit and not strictly bigger than UpperLimit

LimitedCreditEnabled → 1byte

LimitedCreditEnabled	1	-	Encodes if Cmd.LimitedCredit and free Cmd.GetValue are allowed for this file.
	Bit 7-2		RFU
	Bit 1		Free Cmd.GetValue
		0b	No free access to Cmd.GetValue
		1b	Free access to Cmd.GetValue
	Bit 0		Cmd.LimitedCredit support
		0b	Cmd.LimitedCredit disabled
		1b	Cmd.LimitedCredit enabled

Outlet parameters: no

Return value: 0 → success

Non-zero → failure

34.int CreateLinearRecordFile(byte FileNo, byte FileOption, byte[]

AccessRights, byte[] RecordSize, byte[] MaxNoOfRecs);

Function: Create a line record file

Parameter: FileNo → 1bytes

Function: Create a linear record file

Parameter: FileNo → 1 byte

FileNo	1	-	File number of the file to be created.
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5		RFU
	Bit 4-0		File number

FileOption → 1byte

FileOption	1	-	Options for the targeted file.
	Bit 7		Additional Access Rights
		0b	disabled
		1b	enabled
	Bit 6-2		RFU
	Bit 1-0		CommMode (see Table CommunicationModes)
		x0"	CommMode.Plain
		01b	CommMode.MAC
		11b	CommMode.Full

AccessRights → 2bytes

AccessRights	2	-	Set of access conditions for the 1st set in the file (see Setaccessconditions Table).
--------------	---	---	---

RecordSize → 3bytes

MaxNoOfRecs → 3bytes

RecordSize	3	-	Size of a single record in the file.
		0x00001 .. 0xFFFFF	Empty record not allowed.
MaxNoOfRecs	3	-	Maximum number of records within the file.
		0x00001 .. 0xFFFFF	Empty file not allowed.

Outlet parameters: no

Return value: 0→success

Non-zero→ failure

35.int CreateCyclicRecordFile(byte FileNo, byte FileOption, byte[]

AccessRights, byte[] RecordSize, byte[] MaxNoOfRecs);

Function: Create a loop log file

Parameter: FileNo →1byte

Function: Create a loop log file

Parameter: FileNo →1 byte

FileNo	1	-	File number of the file to be created.
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5		RFU
	Bit 4-0		File number

FileOption →1byte

FileOption	1	-	Options for the targeted file.
	Bit 7		Additional Access Rights
		0b	disabled
		1b	enabled
	Bit 6-2		RFU
	Bit 1-0		CommMode (see Table CommunicationModes)
		x0"	CommMode.Plain
		01b	CommMode.MAC
		11b	CommMode.Full

AccessRights →2bytes

AccessRights	2	-	Set of access conditions for the 1st set in the file (see Setaccessconditions Table).
--------------	---	---	---

RecordSize →3bytes

MaxNoOfRecs →3bytes

RecordSize	3	-	Size of a single record in the file.
		0x00001 .. 0xFFFFF	Empty record not allowed.
MaxNoOfRecs	3	-	Maximum number of records within the file.
		0x00002 .. 0xFFFFF	Empty file not allowed; one additional record for backup feature.

Outlet parameters: no

Return value: 0→ success

Non-zero → failure

36.int CreateTransactionMACFile(byte FileNo, byte FileOption, byte[] AccessRights, byte TMKeyOption, byte[] TMKey, byte TMKeyVer);

Function: Create a TransactionMac file

Parameter: FileNo → 1 byte

FileNo	1	-	File number of the file to be created.
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5		RFU
	Bit 4-0		File number

FileOption → 1byte

FileOption	1	-	Options for the targeted file.
	Bit 7-2		RFU
	Bit 1-0		CommMode (see Table CommunicationModes)
		x0b	CommMode.Plain
		01b	CommMode.MAC
		11b	CommMode.Full

AccessRights → 2bytes

AccessRights	2	-	Set of access conditions (see Table TransactionMACAccessRights)
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TMKeyOption → 1bytes

TMKeyOption	1	-	Options for KeyID.AppTransactionMACKey
	Bit 7-2		RFU
	Bit 1-0		KeyType of the KeyID.AppTransactionMACKey
		00b	RFU
		01b	RFU
		10b	KeyType.AES
		11b	RFU

TMKey → 16bytes

TMKeyVer → 1byte

TMKey	16	Full Range	Key value of KeyID.AppTransactionMACKey
TMKeyVer	1	Full Range	Key version of KeyID.AppTransactionMACKey

Outlet parameters: no

Return value: 0 → success

Non-zero → failure

37.int DeleteFile(byte FileNo);

Function: Delete a file

Parameter: FileNo → 1 byte

FileNo	1	-	File number of the file to be created.
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5		RFU
	Bit 4-0		File number
		Full Range	

Outlet parameters: no

Return value: 0→ success

Non-zero →failure

38.int GetFileIDs(byte[] OutData, int[] OutLen);

Function: Get the ID of the file in the application

Parameter: no

Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

39.int GetFileSettings(byte FileNo, byte[] outData, int[] outLen);

Function: Get the configuration of the file

Parameter: FileNo→ byte

FileNo	1	-	File number of the targeted file.
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5		RFU
	Bit 4-0		File number

Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

40.int GetFileCounters(byte FileNo, byte[] outData, int[] outLen);

Function: Get the count count (EV3 proprietary)

Parameter: FileNo→1byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

41.int ChangeFileSettings(byte FileNo, byte FileOption, byte[] AccessRights);

Function: Change the configuration of the file

Parameter:FileNo→1 byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

FileOption →1byte

FileOption	1	-	Options for the targeted file.
	Bit 7		[if not targeting FileType.TransactionMAC] Additional Access Rights
		0b	Not present
		1b	Present
	Bit 7		[if targeting FileType.TransactionMAC] RFU
	Bit 6	-	Secure Dynamic Messaging and Mirroring
		0b	disabled
		1b	enabled
	Bit 5	-	[if targeting FileType.TransactionMAC] TMCLimit configuration
		0b	disabled
		1b	enabled
	Bit 5	0b	[else] RFU
	Bit 4	0b	Reserved
	Bit 3-2	00b	RFU
	Bit 1-0	-	CommMode (see Table CommunicationModes)

AccessRights →2byte

AccessRights	2	-	Set of access conditions for the 1st set in the file (see Setaccessconditions Table).
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Outlet parameters: no

Return value: 0 → success

Non-zero → failure

42.int ChangeFileSettings(byte FileNo, byte FileOption, byte[]

AccessRights, byte[] Option);

Function: Change the configuration of the file

Parameter: FileNo → 1byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

FileOption → 1byte

FileOption	1	-	Options for the targeted file.
	Bit 7		[if not targeting FileType.TransactionMAC] Additional Access Rights
		0b	Not present
		1b	Present
	Bit 7		[if targeting FileType.TransactionMAC] RFU
	Bit 6	-	Secure Dynamic Messaging and Mirroring
		0b	disabled
		1b	enabled
	Bit 5	-	[if targeting FileType.TransactionMAC] TMCLimit configuration
		0b	disabled
		1b	enabled
	Bit 5	0b	[else] RFU
	Bit 4	0b	Reserved
	Bit 3-2	00b	RFU
	Bit 1-0	-	CommMode (see Table CommunicationModes)

AccessRights → 21bytes

AccessRights	2	-	Set of access conditions for the 1st set in the file (see Setaccessconditions Table).
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Option → Optional parameters, please decide according to the parameter description, the following is only partial screenshots

NrAddARs	[1]	0x00 .. 0x07	[Optional, present if Bit7 of FileOption set] Number of additional access conditions sets (not including the 1st and mandatory one) of the targeted file.
AddAccessRights	[NrAddARs * 2]	-	[Optional, present if Bit7 of FileOption set] Sets of access conditions to update the 2nd up to (NrAddARs+1)th set of the file (see Setaccessconditions Table).
SDMOptions	[1]	-	[Optional, present if FileOption[b6] set] SDM Options
	Bit 7	-	VCUID (only for mirroring)
		0b	disabled
		1b	enabled
	Bit 6	-	SDMReadCtr
		0b	disabled
		1b	enabled
	Bit 5	-	SDMReadCtrLimit
		0b	disabled
		'1'	enabled
	Bit 4	-	SDMENCFileData
		0b	disabled
		1b	enabled
	Bit 3-1	000b	RFU
	Bit 0	-	Encoding mode
		1b	ASCII

Outlet parameters: no

Return value: 0 → success

Non-zero → failure

43.int ReadData(byte FileNo, byte[] Offset, byte[] Length, byte

CommMode, byte[] outData, int[] outLen);

Function: Read standard files or backup file data

Parameter: FileNo → 1 byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

Offset → 3bytes

Offset	3	0x000000 .. (FileSize - 1)	Starting position for the read operation.
--------	---	----------------------------	---

Length → 3bytes

Length	3	-	Number of bytes to be read.
		0x000000	Read the entire data file, starting from the position specified in the offset value.
		0x000001 .. (FileSize - Offset)	Read the given amount of bytes.

CommMode→1byte

Communication mode	Bit Representation	Explanation
CommMode.Plain	X0b	No protection: message is transmitted in clear
CommMode.MAC	01b	MAC protection for integrity and authenticity
CommMode.Full	11b	Full protection for integrity, authenticity and confidentiality

Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

44.int WriteData(byte FileNo,byte[] Offset,byte[] Length,byte CommMode,byte[] Data);

Function: Write a standard file or backup file data

Parameter: FileNo→1byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

Offset→3bytes

Offset	3	0x000000 .. (FileSize - 1)	Starting position for the read operation.
--------	---	----------------------------	---

Length →3bytes

Length	3	-	Number of bytes to be read.
		0x000000	Read the entire data file, starting from the position specified in the offset value.
		0x000001 .. (FileSize - Offset)	Read the given amount of bytes.

CommMode→1byte

Communication mode	Bit Representation	Explanation
CommMode.Plain	X0b	No protection: message is transmitted in clear
CommMode.MAC	01b	MAC protection for integrity and authenticity
CommMode.Full	11b	Full protection for integrity, authenticity and confidentiality

Data→ Length byte

Data	X	-	Data to be written.
------	---	---	---------------------

Outlet parameters: no

Return value: 0 → success

Non-zero → failure

45.int GetValue(byte FileNo, byte CommMode, byte[] outData, int[] outLen);

Function: Get the value of the value file

Parameter: FileNo → 1 byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

CommMode → 1 byte

Communication mode	Bit Representation	Explanation
CommMode.Plain	X0b	No protection: message is transmitted in clear
CommMode.MAC	01b	MAC protection for integrity and authenticity
CommMode.Full	11b	Full protection for integrity, authenticity and confidentiality

Established parameters: outData → receives data

outLen → Length of outData

Return value: 0 → success

Non-zero → failure

46.int Credit(byte FileNo, byte CommMode, byte[] Data);

Function: Add value of value files

Parameter: FileNo → 1 byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

CommMode → 1 byte

Communication mode	Bit Representation	Explanation
CommMode.Plain	X0b	No protection: message is transmitted in clear
CommMode.MAC	01b	MAC protection for integrity and authenticity
CommMode.Full	11b	Full protection for integrity, authenticity and confidentiality

Data→4bytes

Data	4	0x00000000 .. 0x7FFFFFFF	Value to be credited. LSB first. Shall be positive.
------	---	--------------------------	---

Parameter: no

Return value: 0→ success

Non-zero→ failure

47.int LimitedCredit(byte FileNo, byte CommMode, byte[] Data);

Function: Limit the value of added value files

Parameter: FileNo→1byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

CommMode→1byte

Communication mode	Bit Representation	Explanation
CommMode.Plain	X0b	No protection: message is transmitted in clear
CommMode.MAC	01b	MAC protection for integrity and authenticity
CommMode.Full	11b	Full protection for integrity, authenticity and confidentiality

Data→4bytes

Data	4	0x00000000 .. 0x7FFFFFFF	Value to be credited. LSB first. Shall be positive.
------	---	--------------------------	---

Parameter: no

return value:0 →success

Non-zero →fail

48.int Debit(byte FileNo,byte CommMode,byte[] Data);

Function: Reduce the value of the value file

Parameter: FileNo→1byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

CommMode→1byte

Communication mode	Bit Representation	Explanation
CommMode.Plain	X0b	No protection: message is transmitted in clear
CommMode.MAC	01b	MAC protection for integrity and authenticity
CommMode.Full	11b	Full protection for integrity, authenticity and confidentiality

Data→4bytes

Data	4	0x00000000 .. 0x7FFFFFFF	Value to be dedited. LSB first. Shall be positive.
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Parameter: no

return value:0 →success

Non-zero →fail

49.int ReadRecords(byte FileNo, byte[] RecNo, byte[] RecCount, byte CommMode, byte[] outData, int[] outLen);

Function: read the data of the record file

Parameter: FileNo→1byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

RecNo →3bytes

RecNo	3	-	Record number. Records are numbered starting with the latest record written.
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		0x000000 .. (ExistingRecCount - 1)	0x000000 meaning the latest record written. The number is limited by the number of existing records (ExistingRecCount).
--	--	------------------------------------	---

RecCount →3bytes

RecCount	3	-	Number of records to be read. Records are always transmitted by the PICC in chronological order (= starting with the oldest, which is Cmd.ReadRecords.RecCount-1 before the one addressed by the given Cmd.ReadRecords.RecNo).
		0x000000	Read all records, from the oldest record up to and including the newest record (given by Cmd.ReadRecords.RecNo) are read
		0x000001 .. (ExistingRecCount - RecNo)	The number is limited by the number of existing records (ExistingRecCount) and the offset given by Cmd.ReadRecords.RecNo.

CommMode→1byte

Communication mode	Bit Representation	Explanation
CommMode.Plain	X0b	No protection: message is transmitted in clear
CommMode.MAC	01b	MAC protection for integrity and authenticity
CommMode.Full	11b	Full protection for integrity, authenticity and confidentiality

Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

50.int WriteRecord(byte FileNo,byte[] Offset,byte[] Length,byte

CommMode,byte[] Data);

Function: Write data to record files

Parameter: FileNo→1byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

Offset →3bytes

Offset	3	0x000000 .. (RecordSize - 1)	Starting position for the write operation within the targeted record.
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Length →3bytes

Length	3	0x000001 .. (RecordSize - Offset)	Number of bytes to be written.
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CommMode→1byte

Communication mode	Bit Representation	Explanation
CommMode.Plain	X0b	No protection: message is transmitted in clear
CommMode.MAC	01b	MAC protection for integrity and authenticity
CommMode.Full	11b	Full protection for integrity, authenticity and confidentiality

Data → Length byte

Data	X	-	Data to be written.
------	---	---	---------------------

Parameter: no

return value: 0 → success

Non-zero → fail

51.int UpdateRecord(byte FileNo,byte[] RecNo,byte[] Offset,byte[] Length,byte CommMode,byte[] Data);

Function: Update the data of the record file

Parameter: FileNo → 1byte

FileNo	1	-	File number of the targeted file.
	Bit 7	-	Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5	00b	RFU
	Bit 4-0	Limited range	File number

RecNo → 3bytes

RecNo	3	-	Record number. Records are numbered starting with the latest record written.
-------	---	---	--

		0x000000 .. (ExistingRecCount - 1)	0x000000 meaning the latest record written. The number is limited by the number of existing records (ExistingRecCount).
--	--	------------------------------------	---

Offset → 3bytes

Length → 3bytes

Offset	3	0x000000 .. (RecordSize - 1)	Starting position for the update operation within the targeted record.
Length	3	0x000001 .. (RecordSize - Offset)	Number of bytes to be updated.

CommMode → 1byte

Communication mode	Bit Representation	Explanation
CommMode.Plain	X0b	No protection: message is transmitted in clear
CommMode.MAC	01b	MAC protection for integrity and authenticity
CommMode.Full	11b	Full protection for integrity, authenticity and confidentiality

Data → Length byte

Data	X	-	Data to be written in the targeted record
------	---	---	---

Parameter: no

return value:0 → success

Non-zero → fail

52.int ClearRecordFile(byte FileNo);

Function: Clear the record file

Parameter: FileNo → 1 byte

FileNo	1	-	File number of the targeted file.
	Bit 7		Second Application Indicator
		0b	Target primary application
		1b	Target secondary application
	Bit 6-5		RFU
	Bit 4-0		File number
		Full Range	

Parameter: no

return value:0 → success

Non-zero → fail

53.int CommitTransaction();

Function: Submit a transaction

Parameter: no

return value:0 → success

Non-zero → fail

54.int CommitTransaction(byte Option, byte[] outData, int[] outLen);

Function: Submit a transaction

Parameter: Option → 1 byte

Option	[1]	-	[Optional] Option parameter
	Bit 7-1		RFU
	Bit 0		Calculated Transaction MAC requested on response
		0b	No TMC and TMV returned
		1b	TMC and TMV returned

Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

55.int AbortTransaction();

Function: Interrupt Transaction

Parameter: no

Parameter: no

return value:0 →success

Non-zero →fail

56.int CommitReaderID(byte[] TMRI, byte[] outData, int[] outLen);

Function: Submit Reader ID

Parameter: TMRI→16bytes

TMRI	16	Full Range	Transaction MAC ReaderID
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Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

57.int Read_Sig(byte Address, byte[] outData, int[] outLen);

Function: Retrieve the ECC original verification signature

Parameter: Address →1byte

Address	1	-	Targeted ECC originality check signature
		0x00	Targeting NXPOriginalitySignature
		0x01 .. 0xFF	RFU

Established parameters: outData→ receives data

outLen→Length of outData

Return value: 0→ success

Non-zero→ failure

7. Demo Test process

